

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

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Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming offers a promising avenue for understanding and potentially treating various pathologies, including spinal muscular atrophy (SMA). This experiment aimed to investigate the efficacy of OSKM transfection in reverting somatic cells to induced pluripotent stem cells (iPSCs) within the context of SMA pathogenesis.

2. Materials and Methods

The experimental procedure involved triggering SMA pathogenesis within a somatic cell population followed by two consecutive OSKM transfections using the Yamanaka factors. Computational analysis was conducted using scVI, performing a total of four analyses to assess cellular identity and population dynamics post-intervention.

Experimental Interventions:

- SMA Pathogenesis Triggered
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

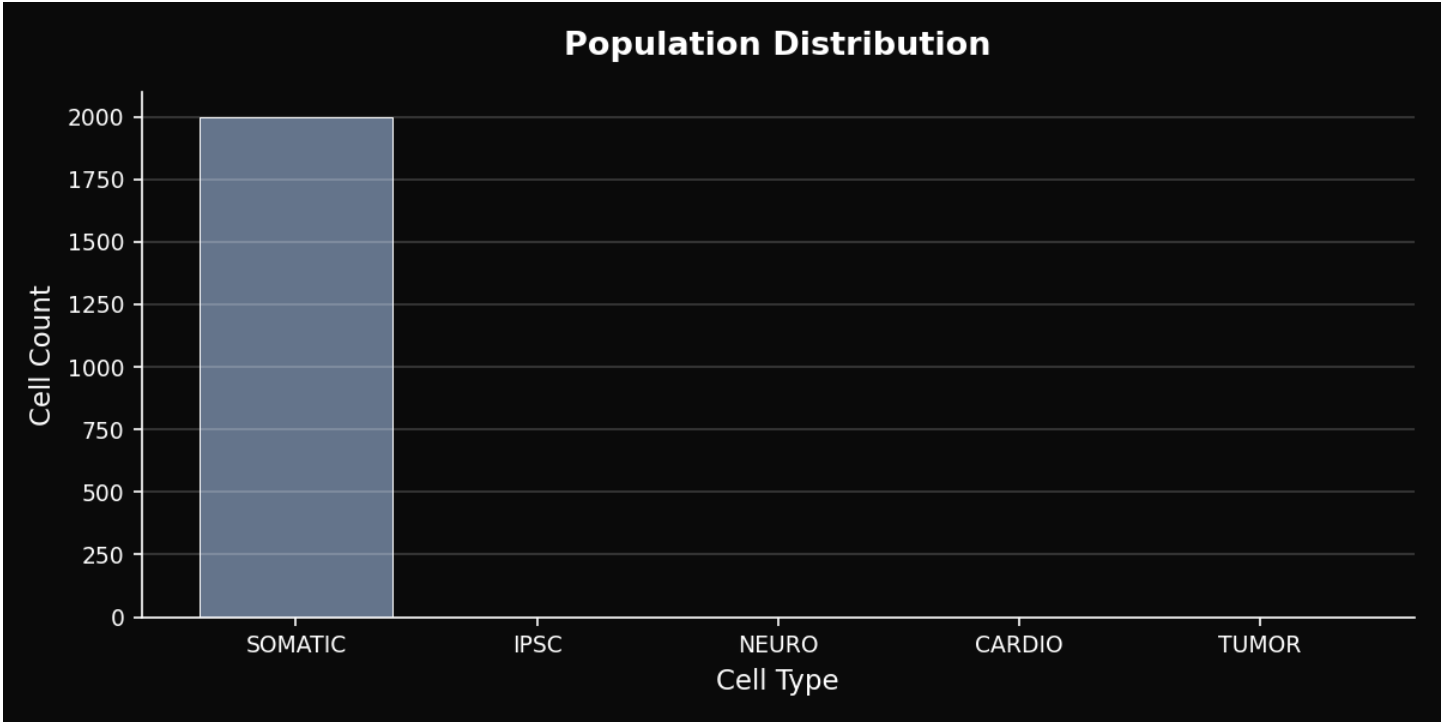
3. Results

Post-intervention analysis revealed a final population consisting solely of 2000 somatic cells, with no detectable iPSC, neurogenic, cardiogenic, or tumorous cells. Despite the dual OSKM transfections, the conversion of somatic cells to iPSCs was not achieved. scVI analyses confirmed the persistence of the somatic cell phenotype across all samples evaluated.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	2000	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 4

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The inability to generate iPSCs from somatic cells under the influence of SMA pathogenesis suggests potential inhibitory effects of this pathological state on cellular reprogramming processes. These findings imply that SMA-induced cellular environments may require additional or alternative reprogramming strategies to overcome inherent barriers to iPSC generation. Future investigations should explore adjunctive factors or modified protocols to enhance reprogramming efficiency under disease-mimicking conditions.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.